UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION 4

Core Physics I — Classical, Quantum and Thermal Physics

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
- **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
- **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
- **Section C**: 4 long-answer questions, 60 marks total.
- An approved calculator is permitted.
- A formula sheet is provided at the end of this paper.
- Show all working for Section C. Marks will be awarded for clear, logical solutions.

Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A simple pendulum of length L has period T. What length of pendulum gives a period of $2T$? [2 marks]

A. $2L$

B. $4L$

C. $\sqrt{2}\,L$

D. $L/2$

2. A simple harmonic oscillator has spring constant $k = 200$ N/m and amplitude $A = 0.1$ m. What is the total energy of the oscillator? [2 marks]

A. 0.5 J

B. 1.0 J

C. 2.0 J

D. 10 J

3. A damped harmonic oscillator has natural angular frequency ω_0 and damping coefficient $\gamma = b/m$. Critical damping occurs when: [2 marks]

A. $\gamma = \omega_0$

B. $\gamma = 2\omega_0$

C. $\gamma = \omega_0/2$

D. $\gamma = \omega_0^2$

4. A forced harmonic oscillator has natural frequency ω_0 and is driven at angular frequency ω. The average power absorbed from the driving force is maximum when: [2 marks]

A. $\omega = \omega_0/\sqrt{2}$

B. $\omega = \omega_0\sqrt{1 - \gamma^2/(2\omega_0^2)}$

C. $\omega = \omega_0$

D. $\omega = 2\omega_0$

5. A transverse wave travels along a string with speed v. If the tension in the string is quadrupled while the linear mass density remains unchanged, the new wave speed is: [2 marks]

A. $v/2$

B. v

C. $2v$

D. $4v$

6. A harmonic wave travels from string 1 (impedance Z_1) to string 2 (impedance $Z_2 = 3Z_1$). The amplitude transmission coefficient t is: [2 marks]

A. 1/2

B. 2/3

C. 3/4

D. 3/2

7. The threshold of hearing is $I_0 = 10^{-12}$ W/m^2. What is the intensity of a sound measured at 60 dB? [2 marks]

A. 10^{-9} W/m^2

B. 10^{-8} W/m^2

C. 10^{-6} W/m^2

D. 10^{-3} W/m^2

8. A stationary source emits sound of frequency $f_0 = 500$ Hz. The sound reflects off a wall that is moving towards the source at speed $v_w = 10$ m/s. Taking the speed of sound in air to be $v = 340$ m/s, the frequency of the reflected sound heard by the source is closest to: [2 marks]

A. 500 Hz

B. 515 Hz

C. 530 Hz

D. 545 Hz

9. An open-open pipe and an open-closed pipe have the same length L. The speed of sound is v. The ratio of the second harmonic frequency of the open-open pipe to the fundamental frequency of the open-closed pipe is: [2 marks]

A. 2

B. 4

C. 8

D. 1

10. Two identical guitar strings are plucked simultaneously. String A is perfectly tuned to 440 Hz. String B has a slightly higher tension than string A. The musician hears beats at a frequency of 3 Hz. Which of the following is true? [2 marks]

A. String B has frequency 437 Hz

B. String B has frequency 443 Hz

C. String B could have frequency 437 Hz or 443 Hz

D. The beat frequency depends on the amplitudes of the two strings

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. In Dirac notation, the object $|\psi\rangle\langle\phi|$ is: [2 marks]

A. A scalar (complex number)

B. A ket (state vector)

C. An operator

D. A bra (dual vector)

2. Let $|\psi\rangle$ be a normalised quantum state and θ a real number. Are the states $e^{i\theta}|\psi\rangle$ and $|\psi\rangle$ physically distinguishable? [2 marks]

A. Yes, they have different norms

B. Yes, they give different measurement probabilities

C. No, they differ only by a global phase and are physically equivalent

D. Only if $\theta = \pi$

3. For two normalised states $|a\rangle$ and $|b\rangle$ in a finite-dimensional Hilbert space, what is $\mathrm{Tr}(|a\rangle\langle b|)$? [2 marks]

A. 1

B. 0

C. $\langle b|a\rangle$

D. $\langle a|b\rangle$

4. For a spin-$\frac{1}{2}$ particle, the commutator $[S_x, S_y]$ is: [2 marks]

A. 0

B. $i\hbar S_z$

C. $\frac{i\hbar}{2} S_z$

D. $2i\hbar\, S_z$

5. A spin-$\frac{1}{2}$ particle is in the state $|\psi\rangle = \dfrac{1}{\sqrt{3}}|{\uparrow_z}\rangle + \sqrt{\dfrac{2}{3}}\,|{\downarrow_z}\rangle$. The probability of measuring S_z and obtaining $-\hbar/2$ is: [2 marks]

A. $1/3$

B. $1/\sqrt{3}$

C. 2/3

D. 1/2

6. A measurement of S_z on a spin-$\frac{1}{2}$ particle yields the result $+\hbar/2$, so the post-measurement state is $|\uparrow_z\rangle$. A subsequent measurement of S_y is immediately performed. The probability of obtaining $+\hbar/2$ for this S_y measurement is: [2 marks]

A. 0

B. 1/4

C. 1/2

D. 1

7. Which of the following statements about stationary states is correct? [2 marks]

A. A stationary state is any normalised quantum state

B. A stationary state is an eigenstate of the Hamiltonian; measurement probabilities in *any* basis are time-independent

C. A stationary state can be a superposition of energy eigenstates with different energies

D. A stationary state has time-independent probabilities only when measured in the energy eigenbasis

8. Two spin-$\frac{1}{2}$ particles form a composite system. The dimension of the composite state space is: [2 marks]

A. 2

B. 3

C. 4

D. 8

9. Two electrons cannot occupy the same quantum state simultaneously. This is a consequence of the fact that electrons are: [2 marks]

A. Bosons, which must be in symmetric states

B. Fermions, which must be in antisymmetric states under exchange

C. Charged particles that repel each other

D. Particles with spin $\frac{1}{2}$ that obey the uncertainty principle

10. In beta-minus decay, a neutron decays as $n \to p + e^- + \bar{\nu}_e$. Which fundamental force mediates this process? [2 marks]

A. The electromagnetic force

B. The strong nuclear force

C. The weak nuclear force

D. Gravity

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1. [15 marks]

Newton's Laws, Friction, and Circular Motion

(a) A block of mass $m_1 = 5$ kg sits on a horizontal surface. The coefficient of static friction is $\mu_s = 0.4$ and the coefficient of kinetic friction is $\mu_k = 0.3$. A horizontal force F is applied to the block.

  (i) What is the maximum horizontal force that can be applied before the block begins to move? 2

  (ii) If a horizontal force $F = 25$ N is applied, find the acceleration of the block. 2

(b) The same 5 kg block is now placed on a *frictionless* inclined plane at angle $\theta = 30°$ to the horizontal. It is connected by a light, inextensible string passing over a frictionless pulley at the top of the incline to a hanging mass $m_2 = 3$ kg. Find the acceleration of the system and the tension in the string. 4

(c) A car of mass $M = 1200$ kg travels at constant speed around a banked curve of radius $r = 80$ m. The banking angle is $\alpha = 20°$ and no friction is required. Find the speed of the car (the "design speed"). 3

(d) A conical pendulum consists of a mass $m = 0.5$ kg on a string of length $L = 1.0$ m. The mass revolves in a horizontal circle, and the string makes a constant angle $\theta = 25°$ with the vertical. Find the tension in the string, the radius of the circular orbit, and the period of revolution. 4

Question 2. [15 marks]

Maxwell–Boltzmann, Lennard-Jones, and Crystal Structure

(a) The Maxwell–Boltzmann speed distribution gives three characteristic speeds for molecules of mass m at temperature T. State the formulae for: the most probable speed c_{mp}, the mean speed c_{avg}, and the root-mean-square speed c_{rms}. (These are on the formula sheet.) 3

(b) Calculate all three speeds for nitrogen molecules (N_2, molar mass $M = 28$ g/mol) at $T = 300$ K. Verify that $c_{\text{mp}} < c_{\text{avg}} < c_{\text{rms}}$. 4

(c) The Lennard-Jones potential is given by

$$V(r) = 4\varepsilon\left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^{6}\right].$$

By differentiating and setting $\mathrm{d}V/\mathrm{d}r = 0$, find the equilibrium separation r_{eq} in terms of a. Hence find the depth of the potential well (the binding energy $|V(r_{\text{eq}})|$) in terms of ε. 4

(d) Iron has a body-centred cubic (BCC) crystal structure with lattice parameter $a_{\text{latt}} = 2.87 \times 10^{-10}$ m. The atomic mass of iron is 55.85 u, where 1 u $= 1.661 \times 10^{-27}$ kg. A BCC unit cell contains 2 atoms. Calculate the density of iron. 4

Question 3. [15 marks]

SHM, Superposition, and Organ Pipes

(a) A block of mass $m = 0.4$ kg is attached to a horizontal spring with spring constant $k = 160$ N/m on a frictionless surface. The block is pulled 8 cm from equilibrium and released from rest. Write the equation $x(t)$ for the subsequent motion. Find the maximum velocity and the maximum acceleration. 3

(b) The same block–spring system is now given an initial displacement $x_0 = 6$ cm and an initial velocity $v_0 = 120$ cm/s in the $+x$ direction at $t = 0$. Find the amplitude A and phase constant ϕ of the resulting oscillation, and write the equation $x(t)$. 4

(c) Two harmonic waves on the same string have the same amplitude $A = 3$ mm and frequency $f = 200$ Hz, but differ in phase by $\delta = \pi/3$. Both waves travel in the $+x$ direction with speed $v = 100$ m/s. Using phasor addition (or the superposition formula), find the resultant amplitude and write the equation for the combined wave. 4

(d) A pipe of length $L = 1.7$ m is open at both ends. The speed of sound is $v = 340$ m/s.

- (i) Find the fundamental frequency and the first three harmonics (i.e. f_1, f_2, f_3, f_4). 2
- (ii) The pipe is now closed at one end. Find the new fundamental frequency and the first three resonant frequencies. 2

Question 4. [15 marks]

Time Evolution and Multi-Particle States

(a) A spin-$\frac{1}{2}$ particle has Hamiltonian

$$H = \omega_0 \, S_z = \frac{\omega_0 \hbar}{2} \left(|\!\uparrow\rangle\langle\uparrow\!| - |\!\downarrow\rangle\langle\downarrow\!| \right).$$

At $t = 0$ the particle is prepared in the state

$$|\psi(0)\rangle = \cos\alpha \; |\!\uparrow\rangle + \sin\alpha \; |\!\downarrow\rangle , \qquad \alpha = \frac{\pi}{6}.$$

Find $|\psi(t)\rangle$. Calculate the probabilities $P(\uparrow_z,\, t)$ and $P(\downarrow_z,\, t)$ and comment on their time dependence. 4

(b) For the state in part (a), calculate the expectation value $\langle S_x \rangle \, (t)$. At what times is $\langle S_x \rangle = 0$? 4

(c) Consider two *distinguishable* spin-$\frac{1}{2}$ particles. Write down a complete orthonormal basis for the composite system. How many basis states are there? Write down one example of an entangled state in this basis. 3

(d) Two identical bosons can each be in one of three single-particle states $|a\rangle$, $|b\rangle$, $|c\rangle$. List all possible two-boson states. How many are there? Repeat for two identical fermions. 4

Formula Sheet

Mathematical Formulae

Binomial expansion: $$(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots \quad (|x| \ll 1)$$

Euler's formula: $$e^{i\theta} = \cos\theta + i\sin\theta$$

Trig identities: $$\sin A + \sin B = 2\cos\left(\tfrac{A-B}{2}\right)\sin\left(\tfrac{A+B}{2}\right)$$ $$\cos A + \cos B = 2\cos\left(\tfrac{A-B}{2}\right)\cos\left(\tfrac{A+B}{2}\right)$$

Mechanics

SUVAT: $$v = u + at, \quad s = ut + \tfrac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \tfrac{1}{2}(u+v)t$$

Dot product: $$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$$

Cross product: $$\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\,\hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

Rocket equation: $$\Delta v = u\ln\left(\frac{m_0}{m_f}\right)$$

Gravitational PE: $$U = -\frac{GMm}{r}$$

Escape velocity: $$v_e = \sqrt{\frac{2GM}{R}}$$

Force from PE: $$F = -\frac{dU}{dx}$$

Elastic collision (1D): $$v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$$

Centre of mass: $$\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$$

Reduced mass: $$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Centripetal accel: $$a_c = \frac{v^2}{r} = \omega^2 r$$

Parallel axis thm: $$I = I_{\text{cm}} + Md^2$$

Atwood machine: $$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L, about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L, about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R \quad \text{(per mole)}$

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6\right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_BT}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_BT}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_BT}{m}}$

Oscillations and Waves

SHM: $x(t) = A\cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$

Damped SHM: $x(t) = Ae^{-\gamma t/2}\cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$

Quality factor: $Q = \omega_0/\gamma$

Forced amplitude: $$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}}$$

Phase lag: $$\tan\delta = \frac{\gamma\omega}{\omega_0^2 - \omega^2}$$

Pendulums: $T_{\text{simple}} = 2\pi\sqrt{L/g}, \quad T_{\text{physical}} = 2\pi\sqrt{I/(Mgh)}$

Wave equation: $$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

String speed: $v = \sqrt{T/\mu}$

Sound in gas: $v = \sqrt{\gamma P/\rho}$

Harmonic wave: $y = A\sin(kx - \omega t)$

String impedance: $Z = \mu v = \sqrt{\mu T}$

Reflection: $$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$$

Power (string): $\langle P \rangle = \frac{1}{2}\mu\omega^2 A^2 v$

Decibels: $\beta = 10\log_{10}(I/I_0), \quad I_0 = 10^{-12}\ \text{W/m}^2$

Doppler: $$f' = f\frac{v \pm v_o}{v \mp v_s}$$

Group velocity: $$v_g = \frac{d\omega}{dk}$$

String harmonics: $$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \ldots)$$

Open-closed pipe: $$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \ldots)$$

Quantum Physics

Schrödinger eq: $$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

Time evolution: $$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar}|E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$$

Spin operators: $$S_z = \frac{\hbar}{2}\big(|{\Uparrow}\rangle\langle{\Uparrow}| - |{\Downarrow}\rangle\langle{\Downarrow}|\big)$$
$$S_x = \frac{\hbar}{2}\big(|{\Uparrow}\rangle\langle{\Downarrow}| + |{\Downarrow}\rangle\langle{\Uparrow}|\big)$$
$$S_y = \frac{\hbar}{2}\big(-i\,|{\Uparrow}\rangle\langle{\Downarrow}| + i\,|{\Downarrow}\rangle\langle{\Uparrow}|\big)$$

Spin eigenstates: $$|{\Uparrow_x}\rangle = \tfrac{1}{\sqrt{2}}\big(|{\Uparrow_z}\rangle + |{\Downarrow_z}\rangle\big), \quad |{\Downarrow_x}\rangle = \tfrac{1}{\sqrt{2}}\big(|{\Uparrow_z}\rangle - |{\Downarrow_z}\rangle\big)$$
$$|{\Uparrow_y}\rangle = \tfrac{1}{\sqrt{2}}\big(|{\Uparrow_z}\rangle + i\,|{\Downarrow_z}\rangle\big), \quad |{\Downarrow_y}\rangle = \tfrac{1}{\sqrt{2}}\big(|{\Uparrow_z}\rangle - i\,|{\Downarrow_z}\rangle\big)$$

Pauli matrices: $$\sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^{8} m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol^{-1}
Gravitational constant	G	6.674×10^{-11} N·m^2/kg^2
Acceleration due to gravity	g	9.81 m/s^2
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^{6} m